

THE ACTION PLAN

- The NRF will be a unique PPP entity, forging collaborations among the industry, academia and govt departments and research institutions
- It will entail an investment of Rs 50,000 cr. during 2023-28, of which Rs 36,000 cr. will come from private partners
- India spends 0.7% of its GDP on R&D compared to the US's 2.8%, China's 2.1% and Israel's 4.9%
- Experts say several disciplines have to work together. Also, a culture for allocation of resources is still evolving in India

report, India has only 253 scientists or researchers per million people.

Beyond the NRF float, fund flow will continue to be the primary concern. "Research funding in India needs major improvements... Participation of private funding and industry in the functioning and resource-generation is the real change that will make a difference," says Rakesh K. Mishra, director, Tata Institute of Genetics and Society (TIGS), Bengaluru. "This will not only attract more applied research proposals to solve societal problems but also draw non-governmental sources of funds to support research."

For this paradigm shift, the NRF will have to nudge companies to invest in R&D. In technologically advanced countries, the private sector steers the surge in research and innovation. In Israel, these private entities contribute a staggering 88 per cent, while in India, it is just 35 per cent. Per an analysis by the Centre for Technology, Innovation and Economic Research, Pune, in 2021, India's top 10 most profitable non-financial firms spent on an average just 0.3 per cent of sales on R&D, with Tata Consultancy Services topping at 1.4 per cent.

The NRF has also been given the task of creating a policy framework and working on regulatory processes. There are also existential challenges to overcome. "Several disciplines have to work together to develop evidence-informed, context-relevant, resource-optimising, culturally-compatible and equity-promoting solutions. Many sections of society are also needed, across different sectors, to effectively implement those solutions," explains Dr K. Srinath Reddy of the Public Health Foundation of India. He points out that the transformation of primary healthcare calls for a confluence of public health, social and behavioural sciences, management, digital technologies and health economics, besides biomedical sciences. "Funding from individuals, industry, and foundations apart from internal resources have helped advance multi-disciplinary research in eyecare for over three decades in India, but the culture of allocation of resources is still evolving," says Dr Gullapalli N. Rao, founder chairman of the LV Prasad Eye Institute in Hyderabad. He quotes a Stanford University study, which revealed that only nine scientists from India figure among the top 2 per cent scientists in the world.

However, the focus on research and innovation has ensured that the number of patents issued has gone up from 4,000 a year a decade ago to 30,000 due to both policy support and a simplified process of patent issuance. Similarly, the number of trademarks issued is up from 70,000 to 250,000 in a decade. But to build a vibrant R&D ecosystem, the NRF will have to strive for a freer flow of funds to advance autonomous and accountable applied research. ■